



Course Name: *FEDF*
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REAL-TIME ANALYTICS PLATFORM

Hotel Revenue Management System

A real-time analytics platform for optimizing hotel room pricing, automating KPI tracking, and maximizing revenue through data-driven insights.

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01 — PURPOSE

Purpose

To optimize hotel room pricing and maximize revenue using real-time analytics and data-driven insights.

Automated KPI Tracking Continuously monitor ADR, RevPAR, and Occupancy rates without manual intervention.	Streamlined Front-Desk Workflows Responsive reservation systems for fast, accurate guest check-ins and room assignments.	Dynamic Market Pricing Eliminate manual adjustments by deploying structural rate multipliers tied to live occupancy levels.
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02 — PROBLEM STATEMENT

Problem Statement

Traditional hotel management relies on static pricing models that fail to react to real-time occupancy fluctuations and market demand.

Manual KPI Tracking Difficulty tracking volatile metrics like ADR and RevPAR without automation.	Multi-Channel Data Gaps Inefficient synchronization leads to overbookings or missed revenue opportunities.	Slow Response Times Delayed reactions to guest cancellations and immediate check-in trends reduce revenue.
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03 — GOALS

Goals

1. Automated KPI Dashboard Centralized real-time audit of ADR, RevPAR, and Occupancy impact across all channels.	2. Smart Forecasting Process historical records to generate tactical business projections and trend models.	3. Algorithmic Pricing Deploy structural rate multipliers tied to instant system occupancy for maximum profitability.
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Scope

✓ In Scope - Real-time KPI Dashboard tracking ADR, RevPAR, and Occupancy - Instant reservation front-desk form for quick check-ins - Dynamic pricing engine with structural rate multipliers - Multi-channel API distribution pipelines	X Out of Scope - Standalone offline desktop application (fully web/cloud-based only) - Third-party payment gateway integrations (mock/simulation only) - Hardware-level keycard generation systems
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Users

Hotel Administrators / Managers Monitor revenue analysis, analyze booking channels, and configure baseline admin pricing multipliers.	Front-Desk Agents Handle fast guest check-ins, process instant cancellations, and assign vacant inventory directly.	Executive Stakeholders Review automated business projections, trend forecasting models, and macro performance statistics.
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Functional Requirements

1 KPI Monitoring Module Evaluate and display ADR, RevPAR, and live Occupancy rates simultaneously on the main dashboard.	2 Dynamic Pricing Engine Automatically update structural rate multipliers based on current inventory status and room fill ratios.
3 Front-Desk Operations	4 Data Integration Layer

Instant Reservation Form with validation for vacant room selection and primary guest name entries.	Compile persistent live APIs across multi-channel distribution paths to prevent booking overlaps.
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07 — NON-FUNCTIONAL REQUIREMENTS

Non-Functional Requirements

⚡ Performance & Latency Live metric updates must calculate and reflect changes within sub-second intervals upon front-desk actions.	🏠 Scalability Securely manage concurrent distribution pipelines without database locks during high-traffic booking windows.	📱 Usability & UI Design Clear modular architecture with intuitive navigation: Dashboard, Revenue Analysis, Booking Channels, Pricing/Admin, and Logout.
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08 — ASSUMPTIONS

Assumptions

📊 Data Availability The system assumes a reliable stream of baseline historical booking records is available to compute precise trend forecasting models.	🌐 Network Connectivity Front-desk workstations maintain stable internet connectivity to process live API distribution pipelines without synchronization lag.	📦 Inventory Accuracy All active room statuses are updated exclusively through the application platform to maintain a single source of truth for occupancy calculations.
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09 — ACCEPTANCE CRITERIA

Acceptance Criteria

Scenario 1: Live Calculation Accuracy Given a front-desk operator registers a new booking, When the reservation is confirmed,	Scenario 2: Algorithmic Pricing Adjustments Given hotel occupancy surpasses a configured threshold,	Scenario 3: Fault-Tolerant Inputs Given the Instant Reservation Form is submitted,
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Then the Occupancy Rate and corresponding RevPAR values must immediately recalculate on the main view.

When a new query checks room availability,
Then the system must display the dynamically multiplied price tier automatically.

When mandatory parameters like "Primary Guest Name" are missing,
Then the interface must halt submission and present explicit input errors.
